Advanced Statistical Methods II

Practice Problems on Non-linear Regression

1 Setup and notation

- Data: $(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)$, where y is the response and $\boldsymbol{x}$ the predictor.
- **Non-linear regression model.** $Y_i = f(\boldsymbol{x}_i, \boldsymbol{\theta}) + \varepsilon_i$, $i = 1, \ldots, n$, where f is known, $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p$ is unknown, and the ε_i are independent with mean 0 and variance σ^2. The model is *non-linear* when f is non-linear in $\boldsymbol{\theta}$.
- **Non-linear least-squares estimator (NLSE).** $\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} S(\boldsymbol{\theta})$, where $S(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\}^2$.
- **Residuals and Jacobian.** $r_i = y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})$, $\boldsymbol{r} = (r_1, \ldots, r_n)^{\mathsf{T}}$, and $\boldsymbol{J} = \boldsymbol{J}(\boldsymbol{\theta})$ is the $n \times p$ matrix with entries $J_{ij} = \partial f(\boldsymbol{x}_i, \boldsymbol{\theta})/\partial \theta_j$.
- **Gradient and Hessian of S.**

$$\nabla S(\boldsymbol{\theta}) = -2\,\boldsymbol{J}^{\mathsf{T}}\boldsymbol{r}, \qquad \nabla^2 S(\boldsymbol{\theta}) = 2\Big(\boldsymbol{J}^{\mathsf{T}}\boldsymbol{J} - \sum_{i=1}^{n} r_i\, \nabla^2 f(\boldsymbol{x}_i, \boldsymbol{\theta})\Big).$$

- **One parameter** ($p = 1$): write $f_i = f(x_i, \theta)$, $J_i = \partial f_i/\partial\theta$ and $f_i'' = \partial^2 f_i/\partial\theta^2$.
- If the ε_i are $N(0, \sigma^2)$, the NLSE is the maximum-likelihood estimator of $\boldsymbol{\theta}$, and $s^2 = S(\widehat{\boldsymbol{\theta}})/(n-p)$ estimates σ^2.

2 Useful results

All three algorithms below have the form $\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \boldsymbol{\delta}^{(k)}$; they differ in how the step $\boldsymbol{\delta}^{(k)}$ is chosen. Here $\boldsymbol{r}$ and $\boldsymbol{J}$ are evaluated at the current iterate $\boldsymbol{\theta}^{(k)}$.

Tool A (Gauss–Newton update). Linearize the model around the current iterate, $f(\boldsymbol{x}_i, \boldsymbol{\theta}) \approx f(\boldsymbol{x}_i, \boldsymbol{\theta}^{(k)}) + \sum_j J_{ij}(\theta_j - \theta_j^{(k)})$, so that $S(\boldsymbol{\theta}^{(k)} + \boldsymbol{\delta}) \approx \|\boldsymbol{r} - \boldsymbol{J}\boldsymbol{\delta}\|^2$. Minimizing over $\boldsymbol{\delta}$ is an ordinary least-squares problem with "response" $\boldsymbol{r}$ and "design matrix" $\boldsymbol{J}$:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + (\boldsymbol{J}^{\mathsf{T}}\boldsymbol{J})^{-1}\boldsymbol{J}^{\mathsf{T}}\boldsymbol{r}, \qquad \text{and for } p = 1: \quad \theta^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2}.$$

Tool B (Newton update). Minimize the second-order Taylor expansion of S around the current iterate. If $\nabla^2 S$ is positive definite,

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \{\nabla^2 S\}^{-1}\nabla S = \boldsymbol{\theta}^{(k)} + \Big(\boldsymbol{J}^{\mathsf{T}}\boldsymbol{J} - \sum_i r_i \nabla^2 f_i\Big)^{-1}\boldsymbol{J}^{\mathsf{T}}\boldsymbol{r},$$

and for $p = 1$: $\theta^{(k+1)} = \theta^{(k)} + \dfrac{\sum_i J_i r_i}{\sum_i J_i^2 - \sum_i r_i f_i''}$.

Tool C (Fisher scoring). For a log-likelihood $\ell(\boldsymbol{\theta})$, replace the observed information $-\nabla^2 \ell$ in Newton's method by the expected information $\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}\{-\nabla^2 \ell(\boldsymbol{\theta})\}$:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \mathcal{I}(\boldsymbol{\theta}^{(k)})^{-1} \nabla \ell(\boldsymbol{\theta}^{(k)}).$$

In non-linear regression with normal errors this is exactly Gauss–Newton. In a generalized linear model with mean $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$ and $\mathrm{Var}(Y_i) = \phi V(\mu_i)$, it is iteratively reweighted least squares (IRLS):

$$\boldsymbol{\beta}^{(k+1)} = (\boldsymbol{X}^\top \boldsymbol{W} \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{W} \boldsymbol{z}, \qquad z_i = \eta_i + (y_i - \mu_i)\, g'(\mu_i), \qquad W_{ii} = \frac{1}{V(\mu_i)\, g'(\mu_i)^2},$$

all evaluated at $\boldsymbol{\beta}^{(k)}$.

Try to derive these three tools before using them.

3 Models and identifiability

Problem 1. (Linear or not?) For each mean function decide: (i) is the model linear in the parameters (use the Jacobian criteria: the model is linear if and only if every $\partial f / \partial \theta_j$ is free of $\boldsymbol{\theta}$)? (ii) Can a transformation make it linear, and under which error structure? (iii) Is there any other trouble?

(a) $f(x, \boldsymbol{\theta}) = \theta_1 + \theta_2 x + \theta_3 x^2$.

(b) $f(x, \boldsymbol{\theta}) = \theta_1 x^{\theta_2}$, $x > 0$.

(c) $f(x, \boldsymbol{\theta}) = \theta_1 + \theta_2 e^{\theta_3 x}$.

(d) $f(x, \boldsymbol{\theta}) = (\theta_1 + \theta_2 x)^2$.

(e) $f(x, \boldsymbol{\theta}) = \dfrac{\theta_1 x}{\theta_2 + x}$.

Problem 2. (What can the data tell us?) For each model, which parameters (or functions of them) can be estimated? Suggest a fix.

(a) $f(x, \boldsymbol{\theta}) = \theta_1 e^{\theta_2 + \theta_3 x}$.

(b) $f(x, \boldsymbol{\theta}) = \theta_1 (1 - e^{-\theta_2 x})$, with $\theta_1, \theta_2 > 0$, when

  (b1) every observation is taken at the same $x_0 > 0$;

  (b2) observations are taken at two distinct values $0 < x_a < x_b$. Is $\boldsymbol{\theta}$ now identifiable? Does the NLSE always exist?

(c) $f(x, \boldsymbol{\theta}) = \theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x}$.

(d) $f(x, \boldsymbol{\theta}) = \beta \cos(\theta x)$, observed at $x = 1, 2, \ldots, n$.

4 Where do closed forms come from?

Most non-linear least-squares problems need iteration, but closed forms appear in several situations:

1. **A linear model in disguise:** $f(x, \theta) = g(\theta)\, x$ with g one-to-one. Fit $\beta = g(\theta)$ by OLS, then invert g (Problems 3 and 4).
2. **A saturated design:** p parameters and only p distinct x values. Then NLS simply interpolates the group means (Problems 5, 6 and 8).
3. **Conditionally linear parameters:** given the non-linear parameters, the linear ones come from OLS in closed form (Problem 7).
4. **Optimal designs:** the best design points often solve a one-dimensional equation (Problem 14).

And when we must iterate, the update is usually a regression in disguise: Gauss–Newton is OLS of the residuals on $\boldsymbol{J}$ (Problems 3, 10 and 11), and Fisher scoring is weighted OLS of a working response (Problem 9).

Problem 3. (A linear model in disguise) Data: $x = (1, 2, 3)$, $y = (2.1,\ 3.9,\ 6.2)$. Model: $Y_i = \theta^3 x_i + \varepsilon_i$, $\theta \in \mathbb{R}$.

(a) Find $\widehat{\theta}$ in closed form. Does it always exist? Is it unique?

(b) Show that the Gauss–Newton update is

$$\theta^{(k+1)} = \theta^{(k)} + \frac{\widehat{\beta} - \{\theta^{(k)}\}^3}{3\{\theta^{(k)}\}^2},$$

where $\widehat{\beta}$ is the least-squares slope of a line through the origin. It is Newton's method for solving which equation?

(c) Starting from $\theta^{(0)} = 1$, compute two iterations. What happens from $\theta^{(0)} = 0$? From $\theta^{(0)} = -1$?

(d) Show that $\sum_i f_i''(\widehat{\theta})\, r_i(\widehat{\theta}) = 0$ here. What does this imply for the speed of Gauss–Newton (see Problem 11)?

Problem 4. (Closed form, but does it exist?) Let $Y_i = e^\theta + \varepsilon_i$, $i = 1, \dots, n$, with ε_i i.i.d. $N(0, \sigma^2)$.

(a) Find the NLSE $\widehat{\theta}$. When does it exist? Compute the probability that it does *not* exist when $\theta = 0$, $\sigma = 2$, $n = 4$. (You may use $\Phi(-1) = 0.1587$.)

(b) Is $\widehat{\theta}$ unbiased? Use a second-order expansion to approximate its bias and variance.

(c) Evaluate the approximations for $e^\theta = 5$, $\sigma = 2$, $n = 4$. A simulation with 4×10^6 samples gives bias -0.0212 and standard deviation 0.212. Comment.

(d) Compare your variance with the general formula $\sigma^2 (\boldsymbol{J}^\top \boldsymbol{J})^{-1}$.

(e) Is $e^{\widehat{\theta}}$ an unbiased estimator of e^θ? What does this say about bias and parametrization?

Problem 5. (Saturated designs)

(a) Suppose the design has exactly p distinct values $x_1, \dots, x_p$, with n_k observations and mean $\bar{y}_k$ at x_k. Show that

$$S(\boldsymbol{\theta}) = \sum_{k=1}^{p} \sum_{i=1}^{n_k} (y_{ki} - \bar{y}_k)^2 + \sum_{k=1}^{p} n_k \{\bar{y}_k - f(x_k, \boldsymbol{\theta})\}^2,$$

and deduce: if $f(x_k, \boldsymbol{\theta}^*) = \bar{y}_k$ for all k, then $\boldsymbol{\theta}^*$ is an NLSE.

(b) $f(x, \boldsymbol{\theta}) = \theta_1 e^{-\theta_2 x}$, with observations at $x = 0$ and $x = c$ only. Find the NLSE in closed form. When does it exist? Evaluate it for $\bar{y}_0 = 100$, $\bar{y}_c = 40$, $c = 2$ hours.

(c) Michaelis–Menten model $f(x; V_m, K) = V_m x / (K + x)$, using only the Puromycin (treated) runs at $x = 0.02$ (rates 76, 47) and $x = 1.10$ (rates 207, 200). Find $\widehat{V}_m$ and $\widehat{K}$ in closed form.

(d) The Lineweaver–Burk method (OLS of $1/y$ on $1/x$) generally gives biased estimates of V_m and K. Why is it *exact* here?

Problem 6. (The three-point method) Asymptotic regression: $y(x) = \alpha + (y_0 - \alpha) e^{-kx}$, $k > 0$. There are three equally spaced design points x, $x + h$, $x + 2h$, with means $\bar{y}_1, \bar{y}_2, \bar{y}_3$.

(a) Show that $y(x + h) - \alpha = \rho \, \{y(x) - \alpha\}$ with $\rho = e^{-kh}$. Deduce the closed forms

$$\widehat{\rho} = \frac{\bar{y}_3 - \bar{y}_2}{\bar{y}_2 - \bar{y}_1}, \qquad \widehat{k} = -\frac{\log \widehat{\rho}}{h}, \qquad \widehat{\alpha} = \frac{\bar{y}_1 \bar{y}_3 - \bar{y}_2^2}{\bar{y}_1 + \bar{y}_3 - 2\bar{y}_2}.$$

Why is this the NLSE?

(b) A cup of coffee reads 90, 70, 56 °C at $t = 0, 5, 10$ minutes. Estimate the room temperature and the cooling rate, and predict the reading at $t = 15$.

(c) When does the solution exist? Try the biochemical oxygen demand (BOD) data at days 1, 2, 3 (8.3, 10.3, 19.0) and at days 3, 4, 5 (19.0, 16.0, 15.6).

(d) Why is the method still useful when there are many data points?

Problem 7. (Profiling out a linear parameter) Consider $Y_i = \theta_1 g_i(\theta_2) + \varepsilon_i$ with $g_i(\theta_2) = 1 - e^{-\theta_2 x_i}$ (the BOD model). Write $\boldsymbol{g}(\theta_2) = (g_1, \dots, g_n)^\mathsf{T}$ and $\boldsymbol{y} = (y_1, \dots, y_n)^\mathsf{T}$.

(a) For fixed θ_2, find $\widehat{\theta}_1(\theta_2)$ minimizing $S(\theta_1, \theta_2)$.

(b) Show that the profile sum of squares is

$$\tilde{S}(\theta_2) = S(\widehat{\theta}_1(\theta_2), \theta_2) = \|\boldsymbol{y}\|^2 - \frac{(\boldsymbol{y}^\mathsf{T}\boldsymbol{g})^2}{\|\boldsymbol{g}\|^2} = \|\boldsymbol{y}\|^2 \sin^2 \alpha(\theta_2),$$

where $\alpha(\theta_2)$ is the angle between $\boldsymbol{y}$ and $\boldsymbol{g}(\theta_2)$. Interpret $\widehat{\theta}_2$ geometrically.

(c) Show that $\dfrac{d\tilde{S}}{d\theta_2} = \dfrac{\partial S}{\partial \theta_2}\Big|_{\theta_1 = \widehat{\theta}_1(\theta_2)}$, and conclude that minimizing $\tilde{S}$ gives the NLSE.

(d) Why is this useful in practice?

Problem 8. (Logistic regression with a binary covariate) The Challenger O-ring data, with launches split at 65°F:

	flights	O-rings at risk	O-rings distressed
cold (< 65°F)	4	24	5
warm (≥ 65°F)	19	114	4

Model: $\operatorname{logit} \pi = \beta_0 + \beta_1 \cdot \text{cold}$, with binomial counts.

(a) Show that the MLE has a closed form: $\widehat{\pi}_g = k_g / m_g$ in each group g, where k_g is the number distressed and m_g the number at risk. Find $\widehat{\beta}_0$ and $\widehat{\beta}_1$, and interpret $e^{\widehat{\beta}_1}$.

(b) Show that the estimated variance of $\widehat{\beta}_1$ is $\frac{1}{k_c} + \frac{1}{m_c - k_c} + \frac{1}{k_w} + \frac{1}{m_w - k_w}$. Give a 95% interval for the odds ratio ($z_{0.975} = 1.96$).

(c) In what sense is this a saturated design, as in Problem 5?

(d) With temperature as a continuous covariate, the same model gives $\widehat{\beta}_1 = -0.1156$ per °F. What is lost by dichotomizing temperature?

Problem 9. (Poisson regression three ways) $Y_i \sim \text{Poisson}(\mu_i)$ independently, with $\log \mu_i = \eta_i = \boldsymbol{x}_i^{\mathsf{T}} \boldsymbol{\beta}$.

(a) Write down the log-likelihood, the score, the observed information $-\nabla^2 \ell$, and the expected information. Show that the last two coincide.

(b) Show that the Newton (equivalently, Fisher scoring) update is a weighted least-squares fit of a working response z_i on $\boldsymbol{x}_i$, with weights w_i. Find z_i and w_i.

(c) Data: $x = (0, 1, 2)$, $y = (2, 3, 6)$, model $\log \mu = \beta_0 + \beta_1 x$. Starting from $\boldsymbol{\beta}^{(0)} = (\log \bar{y},\, 0)$, compute $\boldsymbol{\beta}^{(1)}$ *by hand*.

(d) Why can there be no spurious local maxima here, unlike in non-linear least squares?

(e) Show that at the MLE, $\sum_i y_i = \sum_i \widehat{\mu}_i$.

Problem 10. (One step by hand) Data: $x = (1, 2)$, $y = (0.62,\ 0.18)$. Model: $f(x, \theta) = e^{-\theta x}$. Start: $\theta^{(0)} = 1$.

(a) Compute f_i, r_i, J_i and f_i'' at $\theta^{(0)}$.

(b) Compute one Gauss–Newton step $\theta^{(1)}_{GN}$.

(c) Compute one Newton step $\theta^{(1)}_{N}$.

(d) The minimizer is $\widehat{\theta} = 0.6489$. Evaluate S at $\theta^{(0)}$, $\theta^{(1)}_{GN}$, $\theta^{(1)}_{N}$ and $\widehat{\theta}$. Which step was better? Explain *why*, using the two quadratic models of S that the methods minimize.

(e) "Newton converges quadratically, so it is always better." Comment.

Problem 11. (How fast is Gauss–Newton?) For a one-parameter model, Gauss–Newton is the iteration $\theta^{(k+1)} = G(\theta^{(k)})$ with

$$G(\theta) = \theta + \frac{\sum_i J_i(\theta)\, r_i(\theta)}{\sum_i J_i(\theta)^2}.$$

(a) Show that $G'(\widehat{\theta}) = \dfrac{\sum_i f_i''(\widehat{\theta})\, r_i(\widehat{\theta})}{\sum_i J_i(\widehat{\theta})^2}$.

(b) Deduce the local rate of convergence. When is it faster than linear?

(c) What does the *sign* of $G'(\widehat{\theta})$ tell you about the iterates?

(d) For the data of Problem 10, compute $G'(\widehat{\theta})$ and compare it with the observed error ratios e_{k+1}/e_k, where $e_k = \theta^{(k)} - \widehat{\theta}$:

$$0.041, \quad -0.083, \quad -0.0895, \quad -0.0890, \quad -0.0890.$$

(e) Can Gauss–Newton fail to converge even when started very close to $\widehat{\theta}$?

5 Inference

Problem 12. (Three tests of one hypothesis) Theophylline data, subject 1 ($n = 11$ observations, $p = 3$ parameters; use the t_8 distribution). The one-compartment fit, parametrized by $(\log k_e, \log k_a, \log CL)$, gives

$$\widehat{\log k_e} = -2.9196, \qquad \mathrm{se}(\widehat{\log k_e}) = 0.1709, \qquad S(\widehat{\boldsymbol{\theta}}) = 4.2860.$$

We test $H_0 : k_e = 0.08$ per hour; equivalently, $\log k_e = \log 0.08$; equivalently, the half-life is $t_{1/2} = \log 2/0.08 = 8.66$ hours. Use $t_{8,0.975} = 2.306$ and $F_{1,8,0.95} = 5.318$.

(a) Compute the Wald t statistic on the $\log k_e$ scale, and state your decision at the 5% level.

(b) Use the delta method to obtain $\widehat{k}_e$, its standard error, and the Wald test on the k_e scale.

(c) Do the same on the half-life scale.

(d) Show the exact identity $t_{\text{half-life scale}} = -\,t_{k_e \text{ scale}} \cdot \widehat{k}_e/k_0$, where $k_0 = 0.08$.

(e) Refitting with k_e fixed at 0.08 gives $S_R = 7.5678$. Compute the extra-sum-of-squares F statistic. Why can its value not depend on the scale?

Problem 13. (Why profile equals Wald in linear models)

(a) In the linear model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ ($\boldsymbol{X}$ of full column rank), show that for every b,

$$\tilde{S}(b) - S(\widehat{\boldsymbol{\beta}}) = \frac{(b - \widehat{\beta}_j)^2}{[(\boldsymbol{X}^\top \boldsymbol{X})^{-1}]_{jj}}, \qquad \text{where } \tilde{S}(b) = \min_{\boldsymbol{\beta}:\, \beta_j = b} S(\boldsymbol{\beta}).$$

Deduce that the profile t function $\tau(b) = \mathrm{sign}(b - \widehat{\beta}_j)\sqrt{\tilde{S}(b) - S(\widehat{\boldsymbol{\beta}})}\,/\,s$ equals $(b - \widehat{\beta}_j)/\,\mathrm{se}(\widehat{\beta}_j)$, a straight line.

(b) In a non-linear model, let $\phi = g(\theta_j)$ with g strictly increasing. Show that the profile interval for ϕ is exactly g(profile interval for θ_j). Is the same true for Wald intervals?

(c) Explain why curvature in the plot of $\tau(\theta_j)$ against θ_j is a diagnostic of non-linearity.

6 Design

Problem 14. (Where should we measure?)

(a) Model $\theta_1 e^{-\theta_2 x}$ ($\theta_1, \theta_2 > 0$), with $n/2$ observations at $x = 0$ and $n/2$ at $x = c > 0$. Show that

$$\operatorname{Var}(\widehat{\theta}_2) \approx \frac{2\sigma^2 \theta_2^2}{n\theta_1^2} \cdot \frac{e^{2u} + 1}{u^2}, \qquad u = \theta_2 c,$$

and that the best c solves $u = 1 + e^{-2u}$. Solve this equation by the fixed-point iteration $u^{(k+1)} = 1 + e^{-2u^{(k)}}$ from $u^{(0)} = 1$ (three or four steps suffice), and express the optimal c in half-lives. Why can we not use this design directly?

(b) Michaelis–Menten $V_m x/(K + x)$, with $n/2$ observations at a and $n/2$ at b, where $0 < a < b$ and b is the largest feasible concentration. Show that $\det(\boldsymbol{J}^\top \boldsymbol{J}) \propto \{a(b-a)/(K+a)^2\}^2$, and that the D-optimal choice (maximizing $\det(\boldsymbol{J}^\top \boldsymbol{J})$) is

$$a^* = \frac{Kb}{2K + b}.$$

Evaluate it for $K = 0.0641$, $b = 1.10$ (Puromycin). What happens as $a \to b$? As $a \to 0$?